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Transformer including first and second windings, and first and second cores, and having structure for preventing core cracking

Abstract

A transformer includes a primary winding, a secondary winding, a first core, and a second core. The primary winding and the secondary winding are inserted into the first core and the second core, and the first core and the second core are disposed to face each other. The first core includes one first core portion, and the second core includes a plurality of second core portions. The first core is a lower core, and the second core is an upper core. The first core and the second core are disposed to face each other with a gap interposed between the first core and the second core. The first core and the second core include a heat-resistant elastic body in the gap, and are disposed to face each other. Each of the second core portions is disposed with a heat-resistant elastic body interposed between the second core portions.

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Background/Summary

CROSS-REFERENCE OF RELATED APPLICATIONS

(1) This application is the U.S. National Phase under 35 U.S.C. § 371 of International Patent Application No. PCT/JP2020/046055, filed on Dec. 10, 2020, which in turn claims the benefit of Japanese Patent Application No. 2019-223839, filed on Dec. 11, 2019, the entire disclosures of which Applications are incorporated by reference herein.

(2) The present disclosure relates to a transformer used in a power converter circuit such as a DC to DC converter, a charging apparatus including the transformer, a power supply apparatus including the transformer, and a method of manufacturing the transformer.

BACKGROUND ART

(3) Conventionally, an in-vehicle charging apparatus for charging a rechargeable battery from a commercial AC power supply is mounted on an electric vehicle or a plug-in hybrid vehicle. For example, Patent Document 1 and Patent Document 2 disclose such a charging apparatus. In this case, Patent Document 1 discloses a transformer in which an E-shaped core is vertically combined. In addition, Patent Document 2 discloses a transformer of an outer iron type in which an inner leg is divided from the center thereof and a heat sink is interposed therebetween.

PRIOR ART DOCUMENTS

Patent Documents

(4) [Patent Document 1] Japanese Patent Laid-open Publication No. JP2013-131540A [Patent Document 2] Japanese Patent No. JP5974833B2

SUMMARY OF THE INVENTION

Problems to be Solved by the Invention

(5) However, in the transformers of Patent Documents 1 and 2, there is such a problem that core cracking occurs due to temperature change.

(6) An object of the present disclosure is to provide a transformer capable of avoiding core cracking even when a temperature change occurs, a charging apparatus including the transformer, and a power supply apparatus including the transformer.

Means for Dissolving the Problems

(7) According to the present disclosure, there is provided a transformer including a primary winding, a secondary winding, a first core, and a second core. The primary winding and the secondary winding are wound into the first core and the second core, and the first core and the second core are disposed to face each other. The first core includes one first core portion, and the second core includes a plurality of second core portions.

Effect of the Invention

(8) The transformer of the present disclosure can avoid core cracking even when a temperature occurs, which can enhance the reliability of the transformer.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a block diagram illustrating a configuration example of an in-vehicle charging apparatus **101** according to an embodiment of the present disclosure.

(2) FIG. 2 is a circuit diagram illustrating a configuration example of an LLC resonant type DC to DC converter **105** in FIG. 1.

(3) FIG. 3 is a perspective view illustrating an appearance of a transformer **206** in

(4) FIG. 2.

(5) FIG. 4A is a longitudinal sectional view taken along the line A-A' in FIG. 3.

(6) FIG. 4B is a transverse sectional view taken along the line B-B' in FIG. 3.

(7) FIG. 5 is a longitudinal sectional view displaying an enlarged distortion generated in a core when a temperature difference occurs between upper and lower cores of a transformer according to a conventional example.

(8) FIG. 6 is a longitudinal sectional view when the transformer **206** in FIG. 1 is fixed with an adhesive **404**.

(9) FIG. 7A is a longitudinal sectional view when the transformer **206** in FIG. 2 is being fixed with an adhesive tape **701**.

(10) FIG. 7B is a longitudinal sectional view when the entire circumference of the transformer **206** in FIG. 2 is fixed with the adhesive tape **701**.

(11) FIG. 7C is a perspective view illustrating an appearance when the entire circumference of the transformer **206** in FIG. 2 is fixed with the adhesive tape **701**.

(12) FIG. 8 is a longitudinal sectional view when core portions **303a** and **303b** which are U-shaped cores of the transformer **206** in FIG. 2 are fixed with the adhesive tape **701** with a heat-resistant elastic body **801** interposed therebetween.

(13) FIG. 9 is a longitudinal sectional view when the core portions **303a** and **303b** which are two U-shaped cores and a lower core **302** which is an E-shaped core of the transformer **206** in FIG. 2 are fixed with the adhesive tape **701** with a heat-resistant elastic body **901** interposed therebetween.

(14) FIG. 10 is a longitudinal sectional view illustrating a configuration example of a transformer **206A** according to modified embodiment 1, the transformer **206A** including one I-shaped core **302A** and two U-shaped cores **303Aa** and **303Ab**.

(15) FIG. 11 is a longitudinal sectional view illustrating a configuration example of a transformer **206B** according to modified embodiment 2, the transformer **206B** including one T-shaped core

302B and two L-shaped cores **303Ba** and **303Bb**.

(16) FIG. **12A** is a longitudinal sectional view illustrating a first process in a manufacturing step of the transformer **206** in FIG. **6**.

(17) FIG. **12B** is a longitudinal sectional view illustrating a second process in the manufacturing step of the transformer **206** in FIG. **6**.

(18) FIG. **12C** is a longitudinal sectional view illustrating a third process in the manufacturing step of the transformer **206** in FIG. **6**.

(19) FIG. **12D** is a longitudinal sectional view illustrating a fourth process in the manufacturing step of the transformer **206** in FIG. **6**.

(20) FIG. **12E** is a longitudinal sectional view illustrating a fifth process in the manufacturing step of the transformer **206** in FIG. **6**.

(21) FIG. **13A** is a longitudinal sectional view illustrating a first process in a manufacturing step of the transformer **206** in FIG. **7C**.

(22) FIG. **13B** is a longitudinal sectional view illustrating a second process in the manufacturing step of the transformer **206** in FIG. **7C**.

(23) FIG. **13C** is a longitudinal sectional view illustrating a third process in the manufacturing step of the transformer **206** in FIG. **7C**.

(24) FIG. **13D** is a longitudinal sectional view illustrating a fourth process in the manufacturing step of the transformer **206** in FIG. **7C**.

(25) FIG. **13E** is a longitudinal sectional view illustrating a fifth process in the manufacturing step of the transformer **206** in FIG. **7C**.

(26) FIG. **14** is a perspective view illustrating an appearance of a configuration example of a transformer **206C** according to modified embodiment 3, the transformer **206C** including a lower core **302** which is one U-shaped core and an upper core **303** which is an E-shaped core including core portions **303a**, **303b**, **303c**, and **303d** obtained by dividing the upper core **303** into four portions.

MODE FOR CARRYING OUT THE INVENTION

(27) Hereinafter, embodiments of the present disclosure will be described with reference to the drawings.

Problem of Comparison Example

(28) Hereinafter, the transformer disclosed in Patent Document 1 is referred to as comparison example 1, the transformer disclosed in Patent Document 2 is referred to as comparison example 2, and the problems of these transformers will be described.

(29) In a transformer in which E-shaped cores are vertically combined as in comparison example 1, the core in the longitudinal direction of the E-shaped core is disposed on the outermost side for a rapid temperature change such as a heat cycle test, and the core expands or contracts according to the thermal expansion coefficient of the core following the temperature change. In this case, the temperature change of the inner leg located inside the core delays because the inner leg has a large volume and is located inside, and core cracking occurs at the center of the core due to delay from expansion or contraction in the longitudinal direction of the E-shaped core, and the reliability decreases.

(30) In addition, when a transformer in which the inner leg is divided from the center is applied to a transformer with built-in leakage in which a gap is arranged in an inner leg as in comparison example 2, the manufacturability deteriorates, the balance of the lower face mainly serving as the cooling face and the adhesion of the connection between the cores of an outer leg are not maintained, the stability of the inductance value and the cooling performance of the core are impaired, and the reliability of the transformer operation decreases.

Embodiments

(31) Embodiments according to the present disclosure that solve the above problems will be described below. However, the configuration described below is merely an example of the present

disclosure, the present disclosure is not limited to the following embodiments, and various modifications can be made according to the design and the like without departing from the technical idea according to the present disclosure even in a case other than these embodiments.

(32) FIG. 1 is a block diagram illustrating a configuration example of an in-vehicle charging apparatus **101** according to an embodiment of the present disclosure. The in-vehicle charging apparatus **101** of FIG. 1 is characterized in that AC power from a commercial AC power supply **102** is converted into DC power and outputted to a rechargeable battery **106**, and a transformer **206** incorporated in a DC to DC converter **105** insulates before and after the conversion.

(33) Referring to FIG. 1, the in-vehicle charging apparatus **101** includes a rectifier and smoothing circuit **103**, a power factor correction circuit (PFC circuit) **104**, and the DC to DC converter **105**. In an electric vehicle or a plug-in hybrid vehicle, the rectifier and smoothing circuit **103** rectifies and smooths AC power from the commercial AC power supply **102** of 100 V or 200 V. Next, the PFC circuit **104** performs power factor correction on the input voltage that is rectified and smoothed, suppresses harmonics, and then outputs the voltage to the DC to DC converter **105**. The DC to DC converter **105** converts the input DC voltage into a DC output voltage according to the battery voltage of the rechargeable battery **106** at the subsequent stage.

(34) FIG. 2 is a circuit diagram illustrating a configuration example of the DC to DC converter **105** in FIG. 1. In the present embodiment, as an example, the DC to DC converter **105** of LLC resonant type widely used for a high-efficiency power supply, such as an industrial switching power supply, an in-vehicle charging apparatus, or a power converter, is used as the DC to DC converter.

(35) Referring to FIG. 2, the LLC resonant type DC to DC converter **105** includes, between input terminals T1 and T2 and output terminals T3 and T4, an inverter circuit **201**, a resonance capacitor **209**, a transformer **206**, a rectifying circuit **210**, a smoothing capacitor **211**, and a control circuit **220** that generates gate signals Sg1 to Sg4 for controlling the operation of the inverter circuit **201**. In this case, the inverter circuit **201** is configured by connecting, for example, N-channel MOS transistors **202** to **205** as switching elements in a bridge format and converts a DC voltage into an AC voltage by turning on or off the MOS transistors **202** to **205** according to the gate signals Sg1 to Sg4. In addition, the transformer **206** includes a leakage inductance **207**, an excitation inductance **208** of a primary winding, and an inductance **212** of a secondary winding.

(36) In the DC to DC converter **105**, the inverter circuit **201** converts an input voltage into an AC voltage by switching the input voltage, and outputs the AC voltage to the rectifying circuit **210** via the resonance capacitor **209** and the transformer **206**. In this case, the output voltage is changed by using a frequency modulation method of changing the switching frequency of the four MOS transistors **202** to **205** using the resonance of the two inductances and one capacitor of the leakage inductance **207** and the excitation inductance **208** of the transformer **206** and the resonance capacitor **209**. Next, the output voltage from the transformer **206** is rectified by the rectifying circuit **210** and smoothed by the smoothing capacitor **211**, and then a DC voltage is outputted.

(37) The DC to DC converter **105** configured as described above can reduce the switching loss with the 0-voltage switching, can reduce the surge current and the voltage with the switching current close to the sine wave, and can reduce the noise.

(38) FIG. 3 is a perspective view illustrating an appearance of the transformer **206** in FIG. 2, FIG. 4A is a longitudinal sectional view taken along the line A-A' in FIG. 3, and FIG. 4B is a transverse sectional view taken along the line B-B' in FIG. 3. The upper, lower, left, and right in FIGS. 3, 4A, and 4B will be described as the upper, lower, left, and right directions, but it is not intended to limit the use form of the transformer **206**. As illustrated in FIG. 3, the manufactured transformer **206** is placed on an air-cooling or water-cooling cooling apparatus **305**.

(39) Referring to FIGS. 3 and 4A, the transformer **206** includes a lower core **302** which is an E-shaped core constituting one core portion, two core portions **303a** and **303b** of an upper core **303** which are U-shaped cores, and a bobbin **301** formed between the lower core **302** and the upper core **303** and inserted into a closed magnetic path region **403** formed by the lower core **302** and the

upper core **303**. In this case, a primary winding **401** and a secondary winding **402** are wound around the grooves of the bobbin **301** such that the central axis of the closed magnetic path region **403** becomes the winding axis. It is noted that in the present embodiment, each core **302**, **303** is constituted of, for example, a ferrite core. In addition, the bobbin **301** is made of an insulator material such as a polyphenylene sulfide resin.

(40) The lower core **302** is constituted of an E-shaped core including two outer legs **302a** and **302b**, one inner leg **302c**, and a bottom face portion **302d** for supporting them. In this case, the inner leg **302c** is disposed so as to be inserted into the closed magnetic path region **403**. It is noted that the closed magnetic path region **403** penetrates in the horizontal direction parallel to a core end face.

(41) The upper core **303** is constituted of a core portion **303a** which is a U-shaped core and a core portion **303b** which is a U-shaped core. The core portion **303a** includes an outer leg **303aa**, an inner leg **303ab**, and a bottom face portion **303ac** for supporting them. In addition, the core portion **303b** includes an outer leg **303ba**, an inner leg **303bb**, and a bottom face portion **303bc** for supporting them.

(42) Each of the core portion **303a** and the core portion **303b** of the upper core **303** is disposed so as to face the lower core **302** as follows.

(43) (1) The end face of the outer leg **303aa** of the core portion **303a** is disposed to face the end face of the outer leg **302a** of the lower core **302**, and the end faces are adhered to each other with an adhesive.

(44) (2) The end face of the outer leg **303ba** of the core portion **303b** is disposed to face the end face of the outer leg **302b** of the lower core **302**, and the end faces are adhered to each other with an adhesive.

(45) (3) The respective end surfaces of the inner leg **303ab** of the core portion **303a** and the inner leg **303bb** of the core portion **303b** are separated with a predetermined gap **405** interposed therebetween and are disposed so as to face the end face of the inner leg **302c** of the lower core **302**.

(46) (4) It is noted that the two core portions **303a** and **303b** are disposed to be separated from each other with a predetermined gap **406** interposed therebetween and face each other in the transverse face direction.

(47) In this case, the inner leg **302c** of the lower core **302** inserted into the closed magnetic path region **403** and the inner legs **303ab** and **303bb** of the two core portions **303a** and **303b** inserted into the closed magnetic path region **403** of the upper core **303** are separated with a gap **405** interposed therebetween and disposed to face each other as described above. By adjusting the gap **405**, the excitation inductance **208** constituting the resonance frequency of the LLC resonant type DC to DC converter **105** can be adjusted. In addition, the leakage inductance **207** is adjusted by the distance between the primary winding **401** and the secondary winding **402**, where the thickness of the bobbin between the primary winding **401** and the secondary winding **402**, the difference in the number of windings of the primary winding **401** constituted of two stages in the upper and lower grooves, the difference in the number of windings of the secondary winding **402** constituted of two stages in the upper and lower grooves, and the like.

(48) Further, it can be understood that the outer legs **303aa**, **303ba** and the inner legs **303ab** and **303bb** protrude downward from the bottom face portions **303ac**, **303bc**, respectively, and the closed magnetic path regions **403** are formed between the outer leg **303aa** and the inner leg **303ab** and between outer leg **303ba** and the inner leg **303bb**, respectively, in the transverse section seen in the upward direction in FIG. 4B. It is noted that the closed magnetic path region **403** is filled with a filler **403A** made of, for example, an insulating material such as a silicone filler, at the end of the manufacturing step of the entire core. At this time, the filler **403A** may also enter the gap **405** and the gap **406**.

(49) It is noted that a lead wire, a bobbin cover for securing an insulation distance between the core and the winding, a mechanism for positioning the bobbin and the core, and the like which are

disposed in a common transformer are not illustrated in the drawings, but they may be appropriately added.

(50) Further, the in-vehicle charging apparatus **101** is provided with the cooling apparatus **305** (FIG. **3**) such as a water-cooling apparatus or an air-cooling apparatus to release heat and cool each component. The cooling apparatus **305** is disposed below the transformer **206** to release heat generated by the transformer **206**. At this time, flatness with the bottom face of the lower core **302** greatly affects heat release property. At this time, normally, when each of the cores **302** and **303** generates heat, the cooling apparatus **305** cools the lower core **302**, the upper core **303** is less likely to release heat as compared with the lower core **302** and thus the upper core **303** is hotter than the lower core **302**, and a temperature difference occurs between the upper and lower cores **302** and **303**.

(51) FIG. **5** is a longitudinal sectional view displaying an enlarged distortion generated in each core **302** and **303** when a temperature difference occurs between the upper and lower cores **302** and **303** of a transformer where the upper core **303** is not divided into two portions according to a conventional example. FIG. **5** illustrates a result obtained by enlarging and displaying the analysis result of the distortion generated in the cores **302** and **303** when a temperature difference occurs between the upper and lower cores **302** and **303** like this in an E-shaped core of the conventional example. In this case, the bobbin **301**, the primary winding **401**, and the secondary winding **402** are not illustrated.

(52) As is clear from FIG. **5**, the upper core **303** is deformed convexly upward by the thermal expansion coefficient of the core **303**, and a tensile stress is applied to the outside. This can also be seen from the fact that the thickness of the inner leg of the upper core **303** which had the same size as the thickness of the inner leg of the lower core **302** at the same temperature is thicker than the thickness of the inner leg of the lower core **302** as indicated by the reference character **303p**. When such a stress is applied, core cracking may occur in the E-shaped core of the conventional example, but the core portion **303a** which is a U-shaped core and the core portion **303b** which is a U-shaped core are separated from each other as in the configuration of the present embodiment, which can release the stress and reduce core cracking.

(53) Many in-vehicle passive components are required to comply with the tests defined by the Automotive Electronics Council (AEC), and one of them is a heat cycle test, and resistance to a rapid temperature change is also required. In the upper and lower E-shaped cores of the conventional example, the temperature change of the inner leg located inside the core delays because the inner leg has a large volume and is located inside, and core cracking may occur at the center of the core due to delay from expansion or contraction in the longitudinal direction of the E-shaped core located outside. However, by dividing the upper core **303** into the two core portions **303a** and **303b** which are U-shaped cores as in the configuration of the present embodiment, the volume of the inner leg decreases, and the heat around the inner leg portion is easily transferred to the inner leg due to the gap between the U-shaped cores, and core cracking is less likely to occur. In addition, even when stress occurs, the stress is reduced by the division at the center, and core cracking is less likely to occur.

(54) In the present embodiment, since the transformer **206** of the LLC resonant type DC to DC converter **105** has the gap **405** (FIG. **4A**), the actual manufacturing method needs to be devised. In particular, when the outer legs of the upper core **303** and the lower core **302** are adhered to each other, for example when an epoxy-based adhesive **404** is used to adhere the outer legs as illustrated in FIG. **4A**, the adhesive **404** may be peeled off due to the gap **405**, or a gap which is not intended when designed may be formed between the outer legs (**303aa** and **302a**) (**303ba** and **302b**) of the upper core **303** and the lower core **302**, and the inductance may decrease from a desired inductance.

(55) FIG. **6** is a longitudinal sectional view when the transformer **206** in FIG. **1** is fixed with the adhesive **404**.

(56) To solve the above problem, it is possible to realize a desired configuration by vertically

inverting the transformer **206** at the time of adhering, disposing the upper core **303** including the core portions **303a** and **303b** which are two U-shaped cores on the upper side, disposing the lower core **302** on the lower side, and fixing the outer legs (**303aa** and **302a**) (**303ba** and **302b**) of the upper core **303** and the lower core **302** with the adhesive **404** as illustrated in FIG. 6.

(57) In addition, since the inner legs of the upper core **303** and the lower core **302** in FIG. 5 are separated by the gap, there is a difference in the inner leg thickness deformation between the upper and lower sides. However, if the inner legs are fixed, stress occurs in the core **302** and **303** against deformation so as not to follow the deformation between the upper and lower cores. That is, it is possible to further relax the stress of the cores by not fixing the outer legs of the upper core **303** and the lower core **302** with the adhesive **404**. Therefore, a method of fixing them with an adhesive tape is conceivable, but if the gap **405** is provided, it may be difficult to wind the adhesive tape in the form of FIG. 4A and the like.

(58) FIG. 7A is a longitudinal sectional view when the transformer **206** in FIG. 2 is fixed with an adhesive tape **701**. FIG. 7B is a longitudinal sectional view when the entire circumference of the transformer **206** in FIG. 2 is fixed with the adhesive tape **701**, and FIG. 7C is a perspective view illustrating an appearance when the entire circumference of the transformer **206** in FIG. 2 is fixed with the adhesive tape **701**. In this case, the adhesive tape **701** is an insulating fixing tape for fixing an electric component, such as Kapton (registered trademark) tape manufactured by DuPont, for example.

(59) To solve the above problem, as illustrated in FIG. 7A, when the fixing is performed with the adhesive tape **701**, the adhesive tape **701** is disposed on a flat face **601a** of a placing table **601**, the upper and lower sides of the transformer **206** are reversed, and the core portions **303a** and **303b** of the upper core **303** which are two U-shaped cores are disposed on the lower side and fixed on the adhesive tape **701**. Next, the bobbin **301** around which the primary winding **401** and the secondary winding **402** are wound is inserted and disposed in the closed magnetic path region **403** of the upper core **303**, and covered with the lower core **302**, and finally the entire core is wound around and fixed with the adhesive tape **701**, which can realize the desired configurations of FIGS. 7B and 7C.

(60) FIG. 8 is a longitudinal sectional view when the core portions **303a** and **303b** which are two U-shaped cores of the transformer **206** of FIG. 2 are fixed with the adhesive tape **701** with a heat-resistant elastic body **801** interposed therebetween.

(61) Although the gap **405** is interposed between the core portions **303a** and **303b** which are two U-shaped cores, the heat-resistant elastic body **801** having a heat insulating action may be interposed in the gap **405**, the transformer **206** is vertically inverted in a manner similar to that of FIG. 7, and the transformer **206** may be fixed with the adhesive tape **701** as illustrated in FIG. 8. As a result, the heat-resistant elastic body **801** absorbs the stress due to the temperature difference between the upper core **303** and the lower core **302**, and it is possible to construct a stable transformer structure. It is noted that the heat-resistant elastic body **801** is, a so-called gap filler having a sheet shape, for example and is made of an insulating resin such as a silicone resin.

(62) FIG. 9 is a longitudinal sectional view when the core portions **303a** and **303b** which are two U-shaped cores and the lower core **302** which is an E-shaped core of the transformer **206** in FIG. 2 are fixed with the adhesive tape **701** with a heat-resistant elastic body **901** interposed therebetween.

(63) Further, as illustrated in FIG. 9, a stable transformer structure can be constructed by interposing the heat-resistant elastic body **901** having a heat insulating action in the gap **405**, vertically inverting the transformer **206** in a manner similar to that of FIG. 7, and fixing the transformer **206** with the adhesive tape **701**. It is noted that the heat-resistant elastic body **901** is, a so-called gap filler having a sheet shape, for example and is made of an insulating resin such as a silicone resin. The heat-resistant elastic body **801** in FIG. 8 and the heat-resistant elastic body **901** in FIG. 9 may be formed together.

(64) As described above, the transformer **206** according to the present embodiment can reduce the

stress on the upper core **303** by dividing the upper core **303** into two portions and constituting the lower core **302** of an E-shaped core. As a result, even when a rapid temperature change occurs, a delay in temperature change of the inner legs is reduced between the U-shaped cores to prevent core cracking, and even when the gap **406** is disposed between the inner legs, it is possible to enhance the adhesion between the cores of the outer legs while maintaining the flatness of the lower face of the core and enhancing the heat release property from the lower face of the core, and to enhance the electrical stability such as inductance and the manufacturability of the transformer. This can enhance the reliability.

Modified Embodiment 1

(65) FIG. **10** is a longitudinal sectional view illustrating a configuration example of a transformer **206A** according to modified embodiment 1, the transformer **206A** including one I-shaped core **302A** and two U-shaped cores **303Aa** and **303Ab**.

(66) The core configuration of the embodiment is constituted of one E-shaped core and two U-shaped cores facing the E-shaped core, but the core configuration may be constituted of one I-shaped core **302A** and U-shaped cores **303Aa** and **303Ab** constituting two core portions and facing the one I-shaped core as in modified embodiment 1 of FIG. **10**. In this case, the U-shaped core **303Aa** includes an outer leg **303Aaa**, an inner leg **303Aab**, and a bottom face portion **303Aac** that supports them. In addition, the U-shaped core **303Ab** includes an outer leg **303Aba**, an inner leg **303Abb**, and a bottom face portion **303Abc** that supports them.

(67) In modified embodiment 1 configured as described above, the heat dissipation and the stress relaxation of the core can be realized in a manner similar to that of the embodiment, and further, since the gap **405** is located on the bottom end face of the inner legs **303Aab**, **303Abb**, a leakage magnetic flux **1001** generated in the gap **405** is far from the primary winding **401** and the secondary winding **402**, the leakage magnetic flux **1001** interlinks the primary winding **401** and the secondary winding **402**, and an eddy current loss to occur can also be reduced.

Modified Embodiment 2

(68) FIG. **11** is a longitudinal sectional view illustrating a configuration example of a transformer **206B** according to modified embodiment 2, the transformer **206B** including one T-shaped core **302B** and two L-shaped cores **303Ba** and **303Bb**.

(69) In modified embodiment 2 of FIG. **11**, the core configuration may be constituted of one T-shaped core **302B** and two L-shaped cores **303Ba** and **303Bb** facing the T-shaped core. In this case, the T-shaped core **302B** includes an inner leg **302Bb** and a bottom face portion **302Ba** that supports the inner leg **302Bb**. In addition, the L-shaped core **303Ba** includes an outer leg **303Bab** and a bottom face portion **303Baa** that supports the outer leg **303Bab**. Further, the L-shaped core **303Bb** includes an outer leg **303Bbb** and a bottom face portion **303Bba** that supports the outer leg **303Bbb**.

(70) In modified embodiment 2 configured as described above has the same effects as that of modified embodiment 1.

(71) Next, a process of the method of manufacturing the transformer **206** of FIGS. **6** and **7C** will be described below.

Process Example 1 of Manufacturing Method

(72) FIGS. **12A** to **12E** are longitudinal sectional views illustrating each process of the manufacturing step of the transformer **206** of FIG. **6**.

(73) As illustrated in FIG. **12A**, the core portions **303a** and **303b** of the upper core **303** which are two U-shaped cores are disposed and fixed on the lower side on the flat face **601a** of the placing table **601**. Next, as illustrated in FIG. **12B**, the bobbin **301** around which the primary winding **401** and the secondary winding **402** are wound is inserted into the closed magnetic path region **403** of the upper core **303**. Further, as illustrated in FIG. **12C**, the adhesive **404** is applied to the end faces of the outer legs **303aa**, **303ba** of the core portions **303a** and **303b**. Next, as illustrated in FIG. **12D**, the lower core **302** is disposed to cover the end faces of the outer legs **303aa**, **303ba** of the core portions **303a** and **303b** such that the end faces of the outer legs **302a**, **302b** of the lower core **302**

face the end faces of the outer legs **303aa**, **303ba**, respectively, and the lower core **302** and the upper core **303** are adhered and fixed. Finally, as illustrated in FIG. **12E**, the transformer **206** to be manufactured is obtained by vertically inverting the transformer that is a combined product in which the lower core **302** and the upper core **303** are combined, thereafter installing the transformer **206** in a predetermined case (not illustrated), pouring and curing the filler **403A**, and filling the closed magnetic path region **403** with the filler **403A** to fix the bobbin **301**. In this case, since the transformer **206** generates heat not only in the upper and lower cores **302**, **303** but also in the windings **401**, **402**, the entire transformer **206** is housed in a predetermined case, and the inside of the case is filled with potting made of silicone resin or the like, to release heat of the winding **401**, **402** and the upper and lower cores **302**, **303** to the cooling apparatus **305** or the like below the transformer through the potting.

Process Example 2 of Manufacturing Method

(74) FIGS. **13A** to **13E** are longitudinal sectional views illustrating each process of the manufacturing step of the transformer **206** of FIG. **7C**.

(75) After the adhesive tape **701** is fixed on the flat face **601a** of the placing table **601** as illustrated in FIG. **13A**, the core portions **303a** and **303b** of the upper core **303** which are two U-shaped cores are disposed and fixed on the lower side on the adhesive tape **701** as illustrated in FIG. **13B**. Next, as illustrated in FIG. **13C**, the bobbin **301** around which the primary winding **401** and the secondary winding **402** are wound is inserted and disposed in the closed magnetic path region **403** of the upper core **303**, and then the lower core **302** is disposed to cover the end faces of the outer legs **303aa** and **303ba** of the core portions **303a** and **303b** such that the end faces of the outer legs **302a**, **302b** of the lower core **302** face the end faces of the outer legs **303aa** and **303ba**. Finally, as illustrated in FIG. **13D**, the transformer **206** to be manufactured is obtained by fixing the outer legs and the outer periphery of the bottom face portion of the combined product in which the lower core **302** and the upper core **303** are combined with the adhesive tape **701**, and then vertically inverting the transformer that is the combined product, thereafter, installing the transformer **206** in a predetermined case (not illustrated), pouring and curing the filler **403A**, and filling the closed magnetic path regions **403** with the filler **403A** to fix the bobbin **301**. In this case, since the transformer **206** generates heat not only in the upper and lower cores **302** and **303** but also in the windings **401** and **402**, the entire transformer **206** is housed in a predetermined case, and the inside of the case is filled with potting made of silicone resin or the like, to release heat of the winding **401** and **402** and the upper and lower cores **302** and **303** to the cooling apparatus **305** or the like below the transformer through the potting.

Modified Embodiment 3

(76) FIG. **14** is a perspective view illustrating an appearance of a configuration example of a transformer **206C** according to modified embodiment 3, the transformer **206C** including a lower core **302** which is one U-shaped core and an upper core **303** which is an E-shaped core including core portions **303a**, **303b**, **303c**, and **303d** obtained by dividing the upper core **303** into four portions.

(77) Referring to FIG. **14**, the upper core **303** is divided into four core portions **303a**, **303b**, **303c**, and **303d** each having a rectangular end face shape. In this case, gaps **406**, **406** are interposed between the core portion **303a** and the core portion **303b** and between the core portion **303c** and the core portion **303d**, respectively. In addition, gaps **407** and **407** are interposed between the core portion **303a** and the core portion **303c** and between the core portion **303b** and the core portion **303d**, respectively.

(78) In the embodiment of FIG. **3**, the gap **406** relaxes the stress with respect to the line A-A' direction. On the other hand, modified embodiment 3 has a specific effect that not only the gaps **406** and **406** relax the stress with respect to the line A-A', but also the gaps **407** and **407** relax the stress with respect to the direction orthogonal to the line A-A' direction.

(79) It is noted that, in modified embodiment 3, the upper core **303** is divided into four portions,

but for example, the upper core **303** may be divided into three portions at intervals of 120 degrees around the central portion of the transformer **206**. Similarly, the upper core **303** may be divided into five or more portions around the central portion of the transformer **206**.

Other Modified Embodiments

(80) In the above embodiment, the primary winding **401** is an upper winding, and the secondary winding **402** is a lower winding. However, the present disclosure is not limited to this, and the same effect can be obtained by interchanging the primary winding **401** and the secondary winding **402** or by disposing the primary winding **401** inside and the secondary winding **402** outside or vice versa with respect to the inner legs.

(81) The primary winding **401** and the secondary winding **402** may be copper wires, litz wires, or triple insulated winding wires (TIW).

INDUSTRIAL APPLICABILITY

(82) The transformers **206**, **206A**, **206B**, and **206C** according to the present disclosure are not limited to the DC to DC converter **105** of the charging apparatus **101** in FIG. **1** that supplies a charging voltage to the rechargeable battery **106** and can also be used for various power supply apparatuses that supply a predetermined power supply voltage to a load.

Claims

1. A transformer comprising: a primary winding; a secondary winding; a bobbin on which both the primary winding and the secondary winding are wound, the primary and secondary windings facing each other in a first direction which is an up and down direction; a first core; and a second core, wherein: the first core includes one first core portion, the second core includes a plurality of second core portions, the first core is a lower core, the second core is an upper core, the first core and the second core are disposed to face each other with a gap interposed between the first core and the second core, the plurality of second core portions are disposed side by side in a second direction perpendicular to the first direction, and the bobbin is disposed to be inserted into the one first core portion and the plurality of second core portions.
2. The transformer as claimed in claim 1, wherein the first core and the second core include a heat-resistant elastic body in the gap, and are disposed to face each other.
3. The transformer as claimed in claim 1, wherein each of the second core portions is disposed with a heat-resistant elastic body interposed between the second core portions.
4. The transformer as claimed in claim 1, wherein the first core and the second core are adhered to each other and disposed to face each other.
5. The transformer as claimed in claim 1, wherein the first core and the second core are fixed to each other by an adhesive tape in a state where the first core and the second core are disposed to face each other.
6. The transformer as claimed in claim 1, further comprising a cooling apparatus provided below the first core.
7. The transformer as claimed in claim 1, wherein the first core portion is an E-shaped core, and wherein each of the second core portions is a U-shaped core.
8. The transformer as claimed in claim 1, wherein the first core portion is an I-shaped core, and wherein each of the second core portions is a U-shaped core.
9. The transformer as claimed in claim 1, wherein the first core portion is a T-shaped core, and wherein each of the second core portions is an L-shaped core.
10. The transformer as claimed in claim 1, wherein the first core and the second core are disposed to face each other in the first direction.
11. A charging apparatus that supplies a charging voltage to a rechargeable battery, the charging apparatus comprising a transformer, wherein the transformer comprises: a primary winding; a secondary winding; a bobbin on which both the primary winding and the secondary winding are

wound, the primary and secondary windings facing each other in a first direction which is an up and down direction; a first core; and a second core, wherein: the first core includes one first core portion, the second core includes a plurality of second core portions, the first core is a lower core, the second core is an upper core, the first core and the second core are disposed to face each other with a gap interposed between the first core and the second core, the plurality of second core portions are disposed side by side in a second direction perpendicular to the first direction, and wherein the bobbin is disposed to be inserted into the one first core portion and the plurality of second core portions.

12. The charging apparatus claimed in claim 11, wherein the first core and the second core are disposed to face each other in the first direction.

13. A power supply apparatus that supplies a power supply voltage to a load, the power supply apparatus comprising a transformer, wherein the transformer comprises: a primary winding; a secondary winding; a bobbin on which both the primary winding and the secondary winding are wound, the primary and secondary windings facing each other in a first direction which is an up and down direction; a first core; and a second core, wherein: the first core includes one first core portion, the second core includes a plurality of second core portions, the first core is a lower core, the second core is an upper core, the first core and the second core are disposed to face each other with a gap interposed between the first core and the second core, the plurality of second core portions are disposed side by side in a second direction perpendicular to the first direction, and wherein the bobbin is disposed to be inserted into the one first core portion and the plurality of second core portions.

14. The power supply apparatus as claimed in claim 13, wherein the first core and the second core are disposed to face each other in the first direction.

15. A method of manufacturing a transformer, the transformer comprising: a primary winding; a secondary winding; a bobbin; a first core; and a second core, wherein: the first core includes one first core portion, the second core includes a plurality of second core portions, the first core is a lower core, the second core is an upper core, and the method comprises steps of: winding both of the primary winding and the secondary winding to face each other in a first direction which is an up and down direction, into the bobbin; disposing the first core and the second core to face each other, with a gap interposed between the first core and the second core, disposing the plurality of second core portions side by side in a second direction perpendicular to the first direction, and disposing the bobbin to be inserted into the one first core portion and the plurality of second core portions.

16. The method of manufacturing a transformer as claimed in claim 15, wherein the step of disposing the first core such that the first core faces the second core includes adhering the first core and the second core to each other and disposing the first core and the second core so that the first core and the second core face each other.

17. The method of manufacturing a transformer as claimed in claim 15, wherein the step of disposing the first core such that the first core faces the second core includes fixing the first core and the second core to each other with an adhesive tape in a state where the first core and the second core are disposed to face each other.

18. The method of manufacturing a transformer as claimed in claim 15, wherein the first core and the second core are disposed to face each other in the first direction.
